Single-Frame Deepfake Detection Using Transfer Learning with Vision Transformers and Explainability

Sarah Wood

Abstract

Deepfake detection systems face a fundamental tension: they must be accurate enough to catch sophisticated manipulations while being interpretable enough for human moderators to trust and act on their predictions. This project develops a single-frame deepfake detection system using Vision Transformers with attention-based explainability, evaluated on the Celeb-DF-v2 benchmark dataset. Three models were trained and compared: an EfficientNet-B0 baseline, a ViT-Base/16, and a tuned ViT with stronger augmentation, label smoothing, and increased model capacity. The tuned ViT achieved the strongest class-balanced performance with a macro F1 of 0.72, AUC-ROC of 0.8409, and real recall of 0.86, improving over the CNN baseline's macro F1 of 0.70 and real recall of 0.52. An ablation experiment confirmed that no single modification drove the improvement, requiring the synergistic combination of all changes. Attention map visualization revealed semantically meaningful detection patterns: the model appears to examine facial boundaries and blending regions for fakes while performing targeted feature verification for real images. Structured failure analysis identified identity-specific error concentration, source-level bias between Celeb-real (79.3% accuracy) and YouTube-real (97.0% accuracy), probability calibration issues, and systematic sensitivity to image brightness and edge density. Robustness testing under JPEG compression, Gaussian noise, and resolution loss showed the model is viable under typical social media conditions but fragile beyond moderate degradation thresholds. The project demonstrates that pairing transformer architectures with explainability produces detection systems that are not only more accurate than CNN baselines but interpretable, enabling human-centered content moderation where a model's reasoning can be visually evaluated rather than blindly trusted.

I. Introduction & Problem Statement

With the introduction of Artificial Intelligence to the general public, comes the ability to create things the average person could not create nor easily create. As it was introduced, the almost limitless potential and power are unleashed for anyone to use, a technology able to create a person’s facsimile in impossible circumstances with no guardrails. If not immediately clear, this is about utilizing Artificial Intelligence to create what is called a deepfake. A deepfake is synthetic media in the medium of images, audio, and video; yet it may be partially synthetic or the whole is synthetic. While possible before and not a new circumstance, with more powerful tools available with little to no guardrails, even existent ones can be hopped easily; and once released, there is no withdrawing it. The major issue that arises here is that a human’s ability to

detect something as a deepfake is going to vary vastly, even AI models will vary vastly. Each, human and AI model, has their own strengths and weaknesses defining one's ability to detect something as a deepfake. In a study by Groh, et al titled, "Deepfake detection by human crowds, machines, and machine-informed crowds" the detection ability in humans is the focal point, "We compare the performance of ordinary human observers with the leading computer vision deepfake detection model and find them similarly accurate, while making different kinds of mistakes." (Groh, et al, 2022). The study goes on to find that when the participants were given the AI model's prediction, EfficientNet-B7, the participants' accuracy improved. But when the model was incorrect the human's judgment got dragged down along with it. Groh et al. recommended that explainable AI is necessary for a human's interaction to evaluate whether the answer was valid by the way the AI got to the answer.

Explainable AI is proving increasingly to be a necessity, rather than the AI black box approach. This project addresses that gap by examining two different deep learning architectures in performance, a Convolutional Neural Network (CNN) and a Vision Transformer (ViT). Then the tuned ViT architecture is enhanced with explainability through attention map visualization and facial regions the model appears to focuses on. These attention maps visualize the facial regions a model examines and fixates on when classifying an image; this can take the form of blended boundaries around certain facial features for deepfakes. By showcasing how it is getting input to output, this allows a human to assess the model's predictions are relevant or if its fixation on a certain pattern allows for false positives or negatives. The black box approach to deepfake detection is insufficient for a real-world deployment where trust and oversight are a

requirement. This project demonstrates that the pairing of deep learning architectures with explainability can produce a detection system that can be accurate and interpretable.

II. Dataset

This capstone utilizes the Celeb-DF-v2 dataset (Li et al., 2020), which consists of 590 Celeb-real, 300 YouTube-real, and 5,639 Celeb-synthesis videos. This dataset is publicly available to access, well documented, and has a challenging benchmark with high-quality deepfakes. In the distribution of this dataset it comes with a text file containing a list of videos intended for the test set, so that the results that come from this dataset can be compared with other published works utilizing Celeb-DF-v2.

To prepare the dataset for a single-frame focus, frames were extracted from the video with a sampling rate of every tenth frame. Those frames were then resized to be 224x224 pixels and the JPEG quality set to 90. To reduce the time to extract the grab() method was used on the frames not being sampled to skip them. Using the official test list, videos were assigned to the test set. The remaining videos were split 85/15 into training and validation. This prevents data leakage where frames from the same video could appear in both the training and testing, which would inflate the model's performance since it would recognize the video rather than generalizable detection features. The splitting between training and validation was performed independently within each of the source folders, thus maintaining consistent class ratios across the split as to avoid a possibility of class imbalance due to how prevalent synthetic videos are in this dataset over the real videos. Once the split and frame extraction has finished the extraction produces: 24,856 real / 170,930 fake frames in the train set; 4,307 real /

30,329 fake frames in the validation set; and 7,116 real / 12,939 fake frames in the test set. This gives us 250,477 total frames at 2.78 GB which cuts down on the space needed to handle and process this data significantly from 50 GB. The consequence of this, however, is that the ratio in comparing training and validation to test is off balance. The train ratio stands at 6.88:1; validation's ratio is 7.04:1; and the test is substantially more balanced at 1.82:1. This difference in the distribution makes the comparison between validation and test metrics less direct, but it is acknowledged through weighted sampling, weighted loss functions, and the use of macro F1 as the primary evaluation metric rather than accuracy.

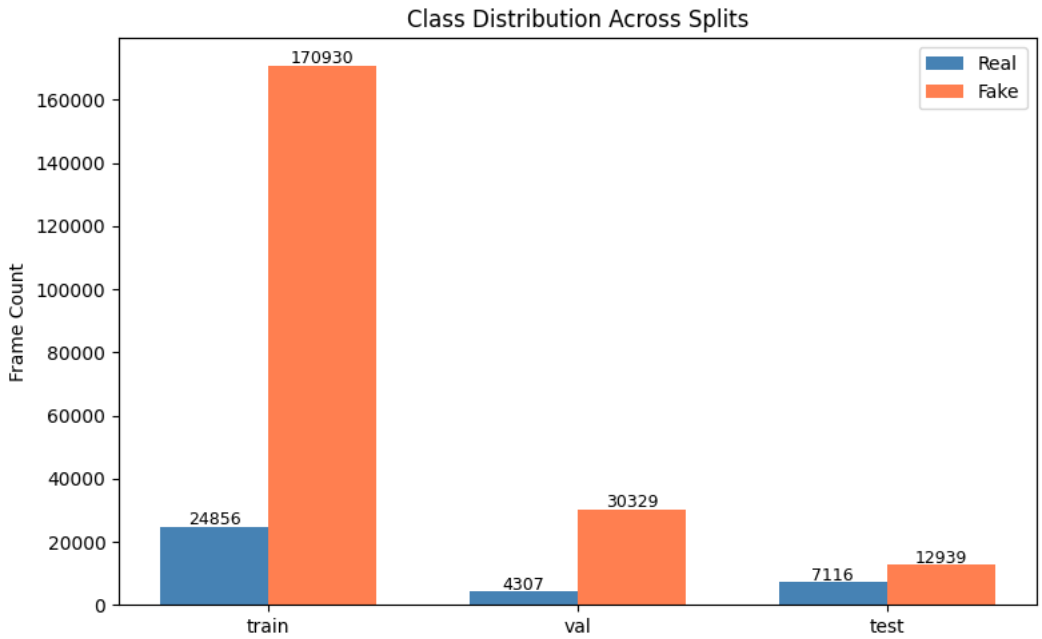

Figure 2-1. Class distribution across train, validation, and test splits

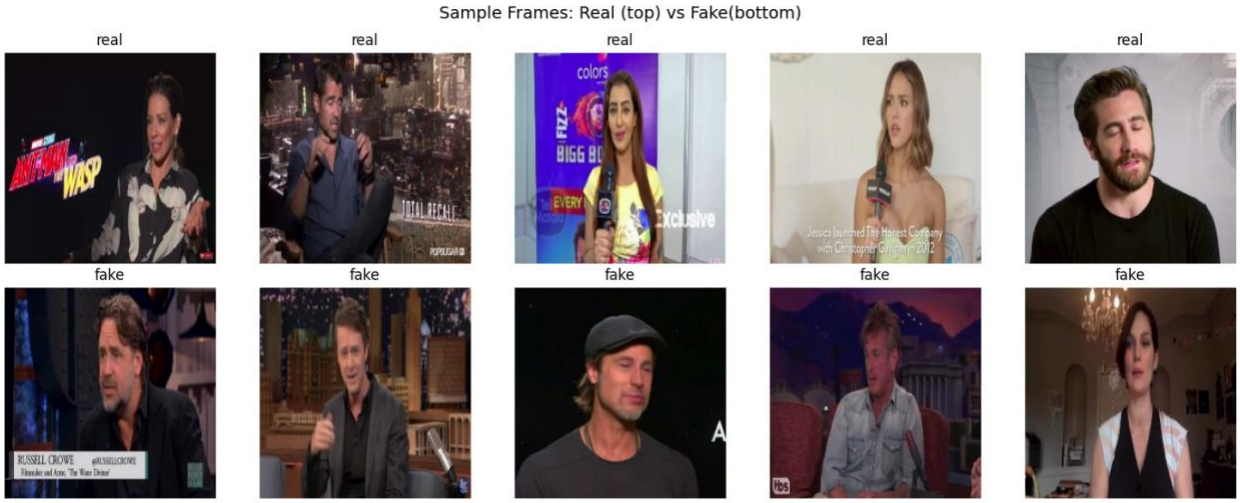

Figure 2-2. Sample frames from the training set. Top: real. Bottom: fake.

III. <u>Baseline: EfficientNet-B0</u>

EfficientNet-B0, a CNN, is the first architecture evaluated as the baseline. EfficientNet-B0 is characterized by its strong performance to parameter ratio which makes it widespread as the baseline in deepfake detection literature. The model is loaded with ImageNet's (Deng et al., 2009) pretrained weights loaded through the timm library. ImageNet is an enormous dataset boasting over 14 million images in thousands of different categories. By loading ImageNet's weights, the model already has generalized visual features like edges, textures, and shapes learned. This transfer learning enables the model to focus on the task at hand, learning real from deepfake instead of starting from the very beginning.

In working with the architecture of EfficientNet-B0, decisions were made regarding its layer architecture with its architecture. This resulted in freezing and unfreezing certain blocks of the CNN in order to avoid accidentally retraining over the early layers. All layers were frozen with exception to the classifier head and the last two

convolutional blocks. The early layers handle the generalized visual features such as edges and textures. The later layers handle the more specific focused features which will be specific to deepfake detection, such as inconsistencies in facial boundaries and blended artifacts. By utilizing freezing and unfreezing this preserves the training from ImageNet and subsequently avoids overfitting as much as possible during the training stage. Another boon to this strategy is that it is computationally more efficient as it lessens the number of parameters that will update during training without the subsequent need for a boost in power to handle those additional parameters. This additionally puts the full focus and computation power at this stage on the latter layers so the one specific task can be addressed: distinguishing real faces versus synthetic ones, making up the bulk of the fine tuning done for this baseline model. The classifier head is class predictions for the model, since there are only two classes the weights need to be adjusted accordingly. This adjustment is needed as the classifier head is set to ImageNet's needs meaning it has over 1,000 categories, so it needs to be decreased to a 2-class output layer for the binary classification: fake (0) and real (1). Through this freezing and unfreezing strategy, only 2,746,142 parameters were trained, making up about 68.5% of the total parameters.

As previously stated with the train and validation sets, there is a significant class imbalance. If training were to go straight on this would affect testing majorly and create a whole host of false positives and negatives. The ratio stands heavily imbalanced in the fake class's favor, in both cases, the train ratio stands at 6.88:1; validation's ratio is 7.04:1. To combat this there are tools available to give the real class more of a chance. First is the WeightedRandomSampler which controls the order and frequency of the

samples in each batch, which to make it pull more from the real class there is a custom sampler being utilized. The sampler takes the inverse of the class frequencies to get a higher weight for that class during training, so it will pay more attention to that class during training. This makes the WeightedRandomSampler oversample that class so the ratio for the batches will be more even. From there the next tool is weighted cross-entropy loss, which compares the model's predicted probability against the label, then produces a number, 0.0 – 1.0, based on how far off the prediction was. The larger this number equates to larger loss, then the model adjusts its parameters to reduce that loss for the next epoch. By weighting this loss not all predictions are penalized equally, since the classes are imbalanced, the model sees a greater number of frames that are synthetic more often which will make it more often predict as the fake class. By weighting the loss, it penalizes the mistake on a real image by about 6.88, this forces the model to take the real class with more attentiveness as it sees those frames less often. Both are needed at different points; the sampler is in the data preprocessing stage when building the model so there is more preventive done at this stage rather than banking on it all in training and fine tuning. By combining both the WeightedRandomSampler and weighted cross-entropy loss, the class imbalance is hit at two different stages: the data itself, and during training. There is one more thing to be with the frames at this stage though, data augmentation. EfficientNet-B0 needs to have normalized RGB channels as it expects the data it gets as an input to be normalized the same as ImageNet was, since that forms its pretrained state. Yet, further things are done to enhance the model's generalization on the frames it receives as its input. With this augmentation besides normalization there are three other augmentations done:

horizontal flips, random rotations, and color jitter. Horizontal flips are applied randomly to the frames with the aim of having the model recognize objects regardless of left or right orientation. Random Rotation rotates the frames at a range of +- 10 degrees, to help the model recognize objects even with slight variance in their orientation. Last there is color jitter, which will randomly change the brightness and contrast of the frames to make the model less sensitive about lighting conditions for an object. These are only applied during training and not validation nor testing as this is the stage in which the model is being evaluated to see how well it is performing and generalizing.

Once the data has been prepared it is time to set up the training and the hyperparameters being utilized there. To make training as efficient as possible, balancing speed with accuracy is a delicate balance, yet there are methods to doing so. Most choices made at this stage are regarding the GPU resource being used, which is T4, it is enough for EfficientNet-B0 and lets us reserve A100 for later points where that extra power is desperately needed. First there are the epochs, which were set to 10, since the model has already been trained, this training serves as more of a fine tuning to the specific task's need. Most of the vital learning the model does is in the first few epochs, which is why early stopping leveraged here so computational resources are not wasted when the model will not improve further. The batch size for each epoch is set to 64, enabling frequent updates to the parameters per epoch, but not so large that the updates eat into potential patterns being missed nor the computational resource not being able to handle that size. This is vital with how many frames are in training, for each epoch it equates to 3,060 parameter updates for every epoch. Making the batch too small has a different effect of making the sampler less effective if it cannot assemble

enough samples for the balance within each batch. For the updates to those parameters, the optimizer is what handles that, for this the Adam optimizer was used. Adam, Adaptive Moment Estimation, in essence tracks each individual parameter and its changes over the recent updates and takes adjustment steps in accordance with those changes. It uses gradients to compute the steps it takes, if it is consistently large the optimizer will have large, confident steps. In contrast if the gradients are noisy and inconsistent, the optimizer adapts to take more small and careful steps. This is advantageous with the pretrained model as the pretrained parameters do not need much alteration, but the classifier head needs more adjustments allowing for the optimizer to fit the needs accordingly. There are more things to control the steps being taken during training, the learning rate is a major one. The learning rate here is set to 1e-4 (0.0001) which is a fine starting point, especially since this is a pretrained CNN, and this training is more about fine-tuning. Here it is small enough that it will not destroy the model's already established patterns, but large enough that it will pick up on the task at hand within the epochs. To further control the learning rate and not have it going at that full rate the entire time the learning rate scheduler comes in. It will watch the validation loss after every epoch to monitor improvement, if it does not have improvement within its given patience, epochs to wait until action, the scheduler will cut the learning rate down. This allows the model to take smaller steps towards the end of its training to settle into the best valley it can find, whereas without it the model will jump around an optimal point without nailing it. Once the training is ended the best model is preserved as seen through the validation AUC-ROC. AUC-ROC is the metric that best fits for gauging the model's performance as purely accuracy with the class imbalance

inflates its performance. AUC-ROC measures how well the model is ranking the classes. This focuses on how often the model assigns a higher fake probability to an actual fake frame than to a real frame, regardless of where the decision threshold is. With this metric, it better reflects the model's true discriminative ability between classes rather than trying to game the majority class.

Once the model has finished training the best model is saved. The checkpoint that it saved at was epoch 5 which resulted in a validation AUC-ROC of 0.7626. After that point the model starts overfitting, and the training loss drops from 0.10 to 0.03 while validation loss rises from 0.75 to over 1.0 at epoch 9. The model starts memorizing the training data at this point and past rather than learning patterns that are generalizable. The learning rate scheduler kicked in at epoch 8 where it cut the rate to 5e-5 (.00005) yet it did not make a meaningful difference as the model became set in its ways. The test set produces a macro F1 of 0.70, AUC-ROC of 0.7705, and an accuracy of 73.93%. By looking at accuracy alone that sounds passable, but the per class breakdown shows more. The model catches fakes well with a recall of 0.86, it identifies 86% of the synthetic frames it encounters, but the real class is where things become interesting. The real class only has a recall of 0.52, almost half of all real frames are being classified as fake. The confusion matrix this shows up as 3,438 real frames as fake out of the 7,116 total real frames in the test set. Real precision sits at 0.67, which means that about a third of the frames the model labels as real are actually fake. If this model was released as is to detect in the wild it would not do well, as it would declare authentic content fake, while at the same time let a third of the deepfakes it says are real slip through without notice.

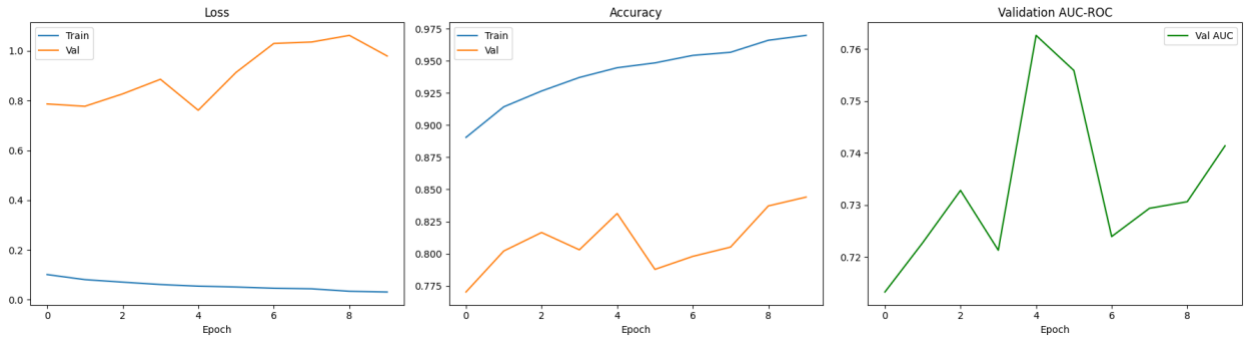

Figure 3-1. EfficientNet-B0 training curves. Validation loss rises after epoch 5, indicating overfitting.

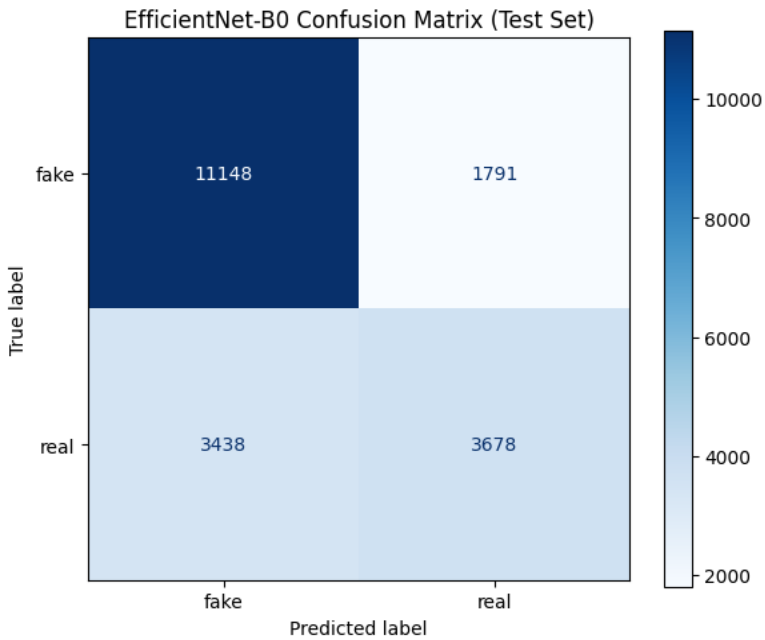

Figure 3-2. EfficientNet-B0 confusion matrix on the test set.

The model's bias towards predicting fake is a direct reflection of the training's class distribution. Despite having the WeightRandomSampler and weighted loss working together to counteract this imbalance, the model is still defaulted to the majority class when uncertain. The gap between fake F1 at 0.81 and real F1 at 0.58 shows that the balancing strategies helped some but were not enough to overcome the 7:1 ratio

through preprocessing and loss weighting alone. The overfitting pattern compounded the problem as the model memorized training specific patterns rather than learning transferable features to help it distinguish real faces with more confidence. These results establish specific targets that the Vision Transformer needs to beat: AUC-ROC above 0.7706, real recall above 0.52, and a more balanced relation between the two classes where one is not being outdone heavily by the other.

IV. Vision Transformer: v1

For the next model being examined and utilized there is the Vision Transformer (ViT), specifically, ViT-Base/16. This is the primary architecture for this project and the model that the explainability layer is built on top of. Where EfficientNet-B0 processed images through convolutional windows that slide across the image to look at small local regions at a time, ViT does things radically different. Here it divides the image into a grid of patches, 16x16 pixels each, which produces 196 patches from the 224x224 frames. Each patch is then treated essentially like a word in a sentence, ViT has origins in Natural Language Processing but was adapted for computer vision, that is then inputted into transformer blocks that use self-attention to weigh how every patch relates to every other patch in the frame simultaneously. This enables a more global view of a frame, which gives it an architectural advantage in this scenario for deepfake detection, as manipulated artifacts will often involve subtle inconsistencies between distant facial regions that a CNN might miss due to its locality. The ViT-Base/16 is loaded with a larger ImageNet, ImageNet-21k, pretrained weights which enriches the foundation of visual features it can build off of.

The freezing strategy is present here and follows a similar logic as the baseline did but is adjusted to fit the transformer architecture. All layers are frozen with the exceptions being later: the classifier head, the final normalization layer, and the last 3 transformer blocks out of the total 12 transformer blocks. This produces a behemoth of trainable parameters; 21.3 million parameters out of 86.6 million in total. This is about 24.6% of the model, which compared to EfficientNet-B0's 68.5%, there is not as much room for ViT to explore which is a more deliberate choice. This is done as transformers are going to be more prone to overfitting when given too many unfrozen parameters to update with a relatively small dataset for its use. So by unfreezing less layers, this enables for ViT to be more manageable as its first pass. The classifier head follows the same 2 class output layer as the baseline did, fake (0) and real (1).

The training set up has several changes from the baseline in order to account for it being a transformer and thus having a different learning style than a CNN. The optimizer is changed from Adam to AdamW, which will decouple weight decay from the gradient update. This change is for a transformer's fine-tuning as it prevents the weight decay from intruding on the adaptive learning rates that Adam is known for. Weight decay is set to a light 0.01 to provide some regularization against overfitting. The learning rate at this point dropped from 1e-4 (0.0001) to 3e-5 (0.00003), which is about a third of the baselines rate. The difference with this is due to ViTs being more sensitive to learning rates and if it were to match the CNN's rate can destabilize the transformer's attention patterns. The scheduler was changed from the ReduceLROnPlateau to a warmup plus a cosine annealing schedule. This is a more dynamic scheduler where instead of waiting for stalling then cutting the rate down; it instead starts the learning rate low then

ramps it up over the first 2 epochs, so the model adjusts gradually. From there it is smoothly decreased over a cosine curve down to 1e-6 (0.000001) at the end. This avoids the sudden jumps that the plateau can cause, which transformers do not handle well. From there, gradient clipping was needed to avoid the gradients becoming massive, it is capped at a max norm of 1.0. Transformers are more vulnerable to exploding gradients than a CNN is. Early stopping was also implemented at a patience set to 5 based on validation's AUC-ROC, thus giving the model more room to explore and find improvements before being stopped. The max epochs in this set up were increased from 10 to 15, since there is the slower warm up period. The class imbalance handling, data augmentation, and batch size are unchanged from the baseline to keep the comparisons fair with our architecture being our isolated variable being changed at this stage. But to accommodate this model being a ViT and having a larger parameter count, the GPU was upgraded from the T4 to the A100, so more accommodations can be made in memory and processing power.

Once training is executed and finished, the best checkpoint ended up at epoch 7 with a validation AUC-ROC of 0.8015, which clears the baseline's AUC-ROC of 0.7706 that it had to beat as its target. After epoch 7 the validation AUC declines across the remaining epochs before early stopping kicked in at epoch 12. The overfitting is still present here but more gradual than it appeared in the baseline. The training accuracy reached 98.8% while the validation accuracy became plateaued around 84-88%, a 12-point gap compared to EfficientNet's 15-point gap. The cosine decay schedule did its job in keeping the model productive longer than the baseline's flat learning rate was able to do. This gave ViT a larger and more useful training window before overfitting

took the main stage. On the test set the model produces: a macro-F1 of 0.72; an AUC-ROC of 0.8137; and an accuracy of 75.77%. Every metric has improved over the baseline's metrics. Going into the classes, the fake recall held at 0.86, while the real recall was able to go from 0.52 in the baseline to 0.58. That stacks to 426 less real frames being flagged as fake, thus dropping the false positives from 3,438 to 3,012. Real precision subsequently improved slightly from 0.67 to 0.69 and fake precision from 0.76 to 0.79. The confusion matrix shows this more cleanly than the baseline did. The ViT ended up at: 11,091 true positives, 4,014 true negatives, 3,012 false positives, and 1,848 false negatives.

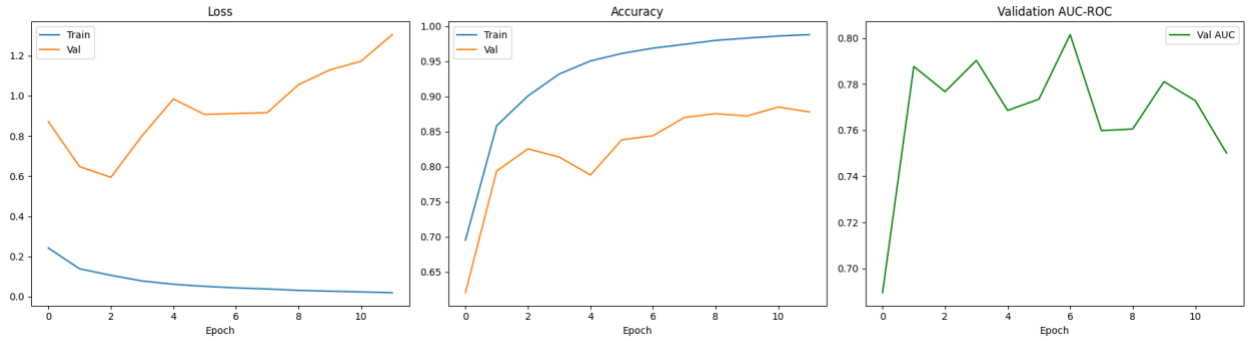

Figure 4-1. ViT v1 training curves. Cosine decay extends productive training.

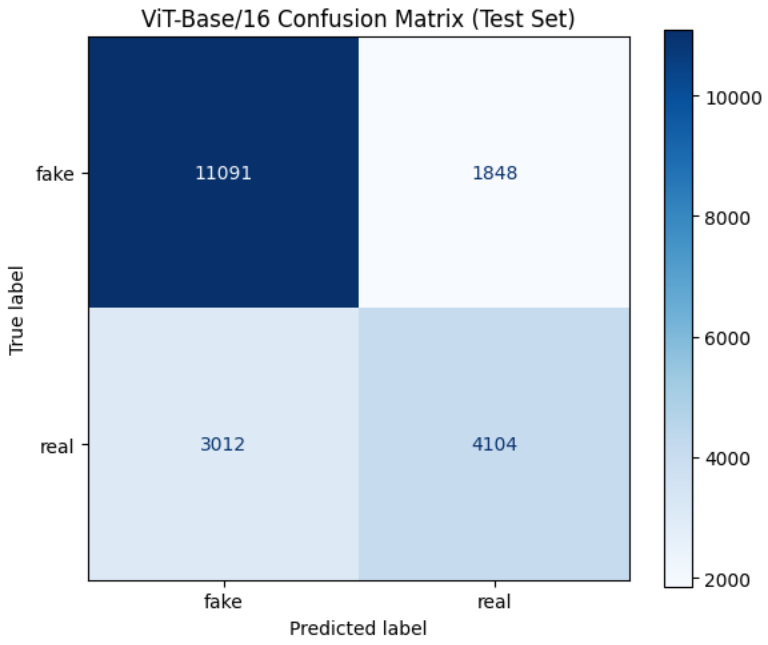

Figure 4-2. ViT v1 confusion matrix on the test set.

By using the transformer architecture, ViT's self-attention mechanism provides a measurable advantage over the CNN for this task, especially in the area where baseline was having the most struggle, correctly identifying real frames. The global attention in ViT allows the model to compare facial regions across the entire image simultaneously rather than relying on local pattern detection, which appears thus to help it distinguish authentic facial consistency from the subtle spatial inconsistencies that deepfake's blending introduces. However, the improvement is merely incremental rather than a transformative experience. Real recall at 0.58 means 42% of real frames are still being misclassified, and the overfitting gap while improved is still substantially existent. The model is better than EfficientNet-B0, but it is not solved. The training dynamics that were changed or introduced means that there is more room for improvement through stronger regularization. Especially since the model has the capacity to memorize training data once cosine stops holding it back. This sets up the tuning experiments in ViT v2, where

the goal will be to push the real recall higher and to close the overfitting gap through more aggressive augmentation, label smoothing, and additional model capacity.

V. Vision Transformer: v2

The second iteration of the ViT takes the initial transformer architecture and targets the two biggest problems from that iteration: overfitting and the real recall. Four changes were made simultaneously in one experiment for the largest impact. First, the data augmentation was strengthened by substantial measures. On top of the existing augmentation, flips, rotations, and color jitter; RandAugment was added which applies two random augmentation operations from a pool of transformations at magnitude 9. Then RandomPerspective was added to a distortion scale of 0.1 applied 30% of the time, which simulates slight camera angle variations. RandomGrayscale was added at 5% probability, forcing the model to occasionally classify without color information. The rotation range was increased from 10 to 15 degrees and color jitter intensity was pushed from 0.2 to 0.3 for brightness and contrast. The intent in doing this is to make training harder on the model so it cannot so easily memorize the specific images and is forced to learn the more generalizable features instead. Second, label smoothing was added on top of the cross-entropy loss at a value 0.1. instead of telling the model that the frame is 100% fake or 100% real, the targets become 95% and 5% respectively. This will discourage the model from overconfidence in its predictions, which in turn acts as another form of regularization. Third, the quantity of transformer blocks that are unfrozen during training was doubled, going from 3 to 6, which increases the number of our trainable parameters from 21.3 million to 42.5 million. This will give the transformer more capacity to learn the deepfake specific features, but it does come at a greater risk

of overfitting, which is why it needs backup from stronger augmentation and label smoothing to counteract this behavior. Lastly, the GradScaler is introduced for ViT v1 to limit exploding gradients is removed, to see if accuracy will be improved, though not given quantification in isolation due to other factors at play concurrently. Hyperparameters were lightly altered here to give ViT v2 more space, but most were untouched: AdamW at 2e-5; a 3-epoch warmup instead of 2-epoch plus cosine schedule; early stopping at patience set to 5; batch size at 64; and the class imbalance handling all remain untouched. Additionally, torch.backends.cudnn.benchmark was set to False to improve training stability and reproducibility.

In training we are met with a complicated story. The best checkpoint kicked in at epoch 2 with the validation AUC-ROC of 0.8103, and after the validation AUC declined across the subsequent epochs before early stopping was triggered at epoch 7. It is concerning that the best model was met at 2/15 epochs, since training runs are quite a bit longer. The learning rate may have been too high for the 42.5 million trainable parameters, or 2 epoch warm up may have been too short for the amount of capacity that was unfrozen, or that the augmentation and label smoothing made the training too noisy for the model to improve after the initial warm up. This is concerning with how the validation accuracy continued to climb while the validation AUC fell from 0.8103 after epoch 2 across the remaining epochs. With the divergence in these two metrics, it signals that the model's probability calibration is degenerating. The model becomes more confident in its predictions, while pushing probabilities towards 0 and 1, but confidence does not mean it is reliably ordered by correctness which is what collapses

the AUC. Label smoothing was supposed to counteract such but proved to not be enough for the learning rate and increase in unfrozen parameters.

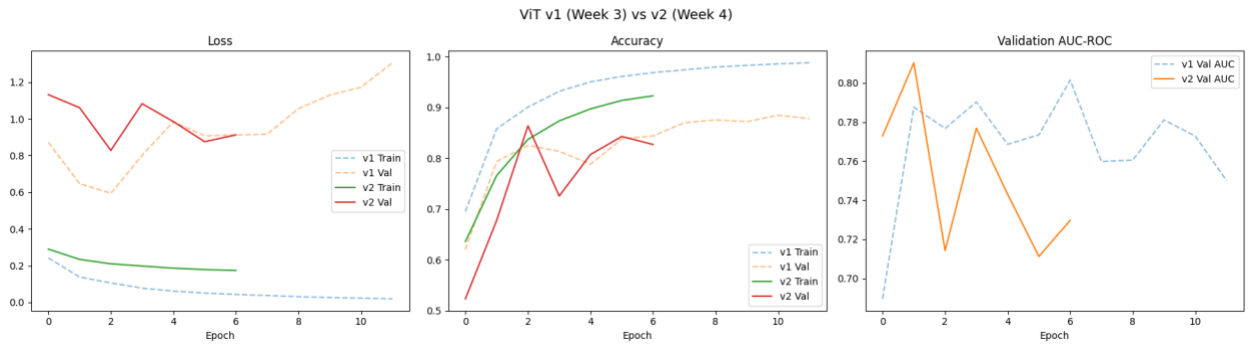

Figure 5-1. Training curve comparison between ViT v1 (dashed) and v2 (solid)

In our test set the model produces a macro-F1 of 0.72, an AUC-ROC of 0.8409, and an accuracy of 72.81%. These were the best metrics produced across all experiments previous. In the class breakdown, the real recall jumped to 0.86 from 0.52 in the baseline, and 0.58 in v1. This was the largest improvement in this project, and subsequently the false positives dropped from 3,438 at the baseline to 968 at v2, giving a 72% reduction. Fewer real frames are being incorrectly flagged as fake. Though the cost of this is the fake recall dropped from 0.86 to 0.65. Which is a 21-point decrease that is significant. False negatives had an increase from 1,791 in the baseline to 4,486 in v2. The model shifted toward real-class detection at the cost of fake detection. The real precision is at 0.58 which is notable: 42% of the frames the model labels as real are in truth, fake. This goes in hand with the fake recall's decrease showcasing a circumstance where four in ten frames would be a missed deepfake in a deployment scenario.

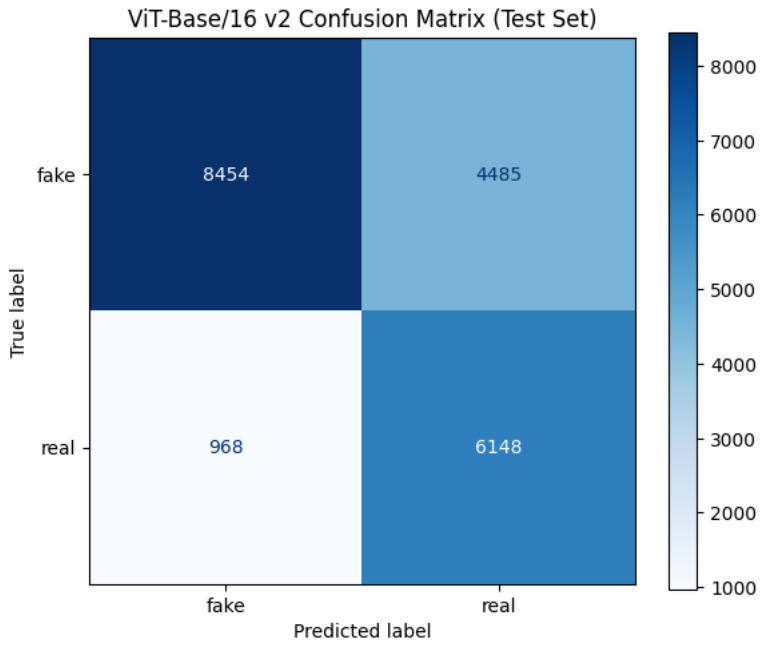

Figure 5-2. ViT v2 confusion matrix on the test set.

There is notable discrepancy between the validation and test metrics that must be addressed separately. The best v2 validation AUC was 0.8103, which clears v1's best validation AUC of 0.8015. In testing v2 also has the higher AUC at 0.8409 versus v1's 0.8148. In this retrain, the validation and test metrics agree directionally, with v2 outperforming v1 on both. The class distribution difference between splits still exists and is noted but does not produce the contradictory signal seen in the original run. This looks like an unassuming improvement but the model that was saved by the validation criterion is weaker in validation than v1. Yet by examining the class distribution there is something there between the splits. The validation is heavily 7:1 fake to real, while the test set is more balanced at 1.8:1. AUC-ROC is theoretically distribution independent, but when the error rates are differing across classes as they do here, the test set being more balanced yields a higher AUC. The jump seen from validation AUC 0.8103 to the

test AUC 0.8409 is not a generalization improvement. It is partially due to the class distribution effect. To be clear this is not invalidating the results the models produce but it means that the test AUC requires more context in interpretation rather than straightforward.

Since four changes were made simultaneously, giving credit to a single modification is impossible without ablation. To address this an ablation experiment was done afterwards. V1's architecture with 3 unfrozen blocks with the original augmentation, with only label smoothing added. If label smoothing alone produces the match to v2's performance, this would make it the primary driver. If not, the capacity increase or augmentation contributed meaningfully to the whole. The ablation here tells a clear path. It achieved a macro F1 of 0.70, which was identical to EfficientNet-B0, an AUC-ROC of 0.8032, and a test accuracy of 70.54%. Real recall shot up to 0.78, which was the highest of any model, but fake recall collapsed to 0.66, which was the lowest of any model. The model swung from predicting mostly fake to mostly predicting real without any learning any better features. This expanded the false negatives to 4,349, comparable to v2's 4,486, though the ablation achieved this through a bias shift rather than genuine learning as evidenced by its collapsed fake recall of 0.66. It seems that label smoothing alone shifted the model's bias without giving it the representation capacity or data diversity to improve its discrimination ability. The v2 improvement required the full combination of stronger augmentation, increased capacity, and label smoothing to work together to produce the meaningful change. Not one by itself was sufficient to get the model going.

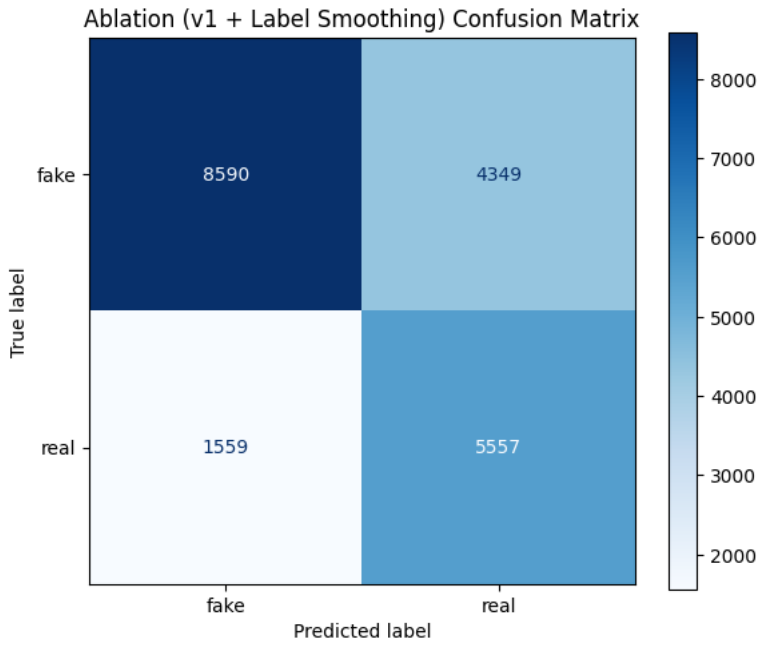

Figure 5-3. Ablation experiment (v1 + label smoothing only) confusion matrix.

With a three-model progression from EfficientNet-B0 to ViT-Base/16/v1 and ViT-Base/16/v2, shows consistent improvement in AUC-ROC from 0.7706 to 0.8148 to 0.8409 and in real recall from 0.52 to 0.58 to 0.86, with macro F1 from 0.70 to 0.72 in both v1 and v2. The ablation confirms that these gains were symbiotic rather than one technique being the carrier of improvement. However, the training dynamics in v2, with the calibration deterioration, the validation/test distribution effect, and the fake recall tradeoff all indicate that the model is improved but not deployable without additional safeguards. The next phase of this project shifts from model improvement to understanding it, using the ViT's attention mechanism to visualize what the model appears to focus on when it predicts and conducting a structured failure analysis to identify where and why it still gets things wrong.

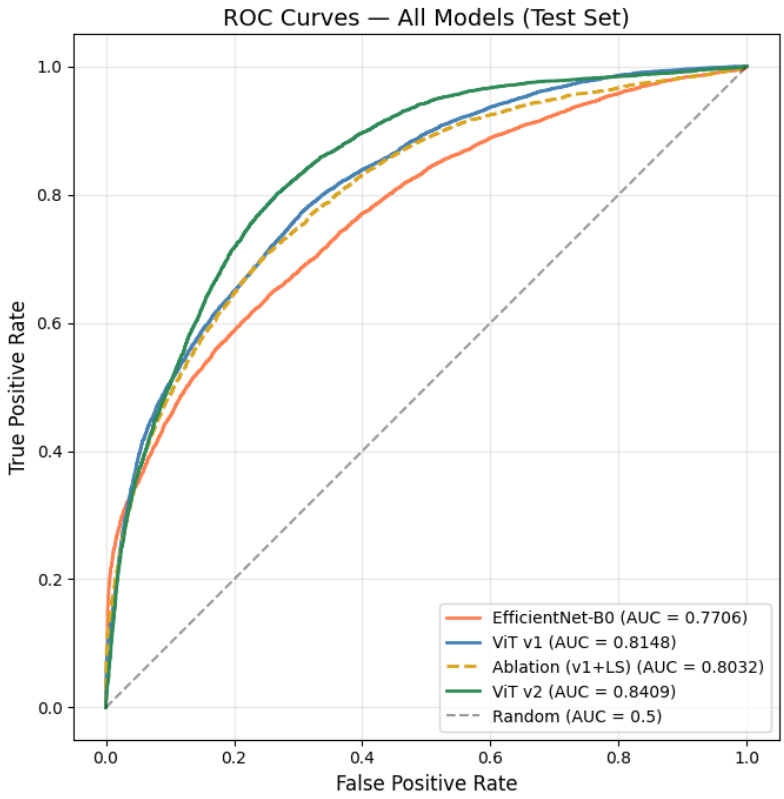

Figure 5-4. ROC curves for all four models on the test set.

VI. <u>Explainability</u>

The ViT's architectural advantage over CNN is not just in performance but what comes after it makes its prediction. While a CNN produces a just a classification, the reasoning behind the prediction is mixed with millions of parameters, lacking a straightforward window to inspection. The ViT's self-attention mechanism provides a built-in window into its reasoning. Every transformer block computes its attention weights that in essence describe how much each patch of the image attends to every other patch. By extracting the CLS (classification) token's attention to all 196 patch tokens from the final transformer block, then averaging across all attention heads, and reshaping back into a 14x14 grid, the result is a heatmap that can show which regions

of the image the model focusing on when making its prediction. This is not so much as a separate explainability tool built into the transformer, but a direct readout of the transformer's own internal processing. This is what makes ViT attention maps more direct about the actual predictions rather than CNNs post-hoc methods such as Grad-CAM.

Attention maps were generated across four prediction categories: true positives, true negatives, false positives, and false negatives. For correctly detected fakes, the attention spreads broadly across the face, concentrating on the eyes, nose bridge, jawline, and facial boundaries. These are regions where deepfake blending artifacts are likely to appear, where the generated meets the original. Rather than fixating on any single feature, the model appears to attend to areas where the synthesis of deepfakes produces spatial inconsistencies. In correctly identified real frames the pattern is different. Attention is focused on a few specific features, such as the chin or mouth area, rather than a broad pattern seen in fake frames. The model in this case appears to verify a small number of facial features for consistency rather than search specifically for artifacts. This difference in apparent attention strategy between the two classes is meaningful and aligns with domain knowledge about where deepfake manipulation is detectable.

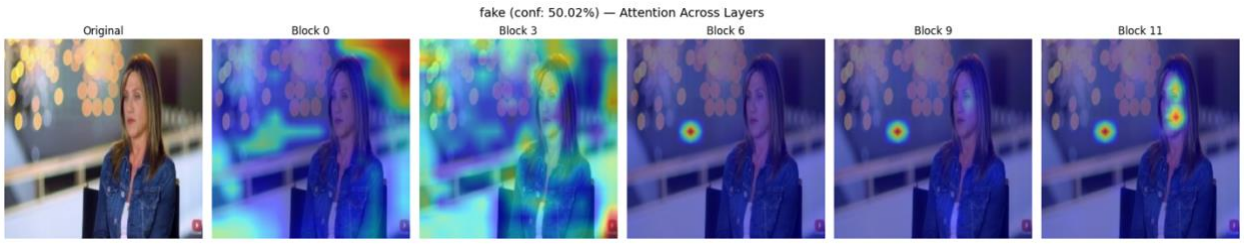

Figure 6-1. Attention map overlay for a correctly detected fake frame.

Figure 6-2. Attention map overlay for a correctly identified real frame.

The failure cases reveal how the model's attention breaks down when the predictions go wrong. The most confident false positives all originate from a single identity, id48, from the Celeb-real folder. This is not a general failure to recognize real frames. It is an identity-specific bias where the model has learned features from this particular individual's face that overlap with its learned fake indicators. More concerning is that the attention maps for these false positives show significant activation on non-facial elements, particularly background text and logos. The model is partially basing its fake prediction on background content rather than facial analysis, a form of spurious correlation where certain background patterns become associated with the fake class during training. For false negatives the attention tells a different story. The most confident missed fakes hover near 50% confidence, meaning the model is essentially taking a guess on the classification. The attention maps show narrowly focused activation on a single small region rather than the distributed facial scanning pattern that characterizes successful fake detection. The model cannot find manipulation artifacts and defaults to looking at one spot without the broader boundary scanning that would reveal the synthesis. These fakes evaded detection because they lack the visible blending artifacts the model appears to have learned to identify.

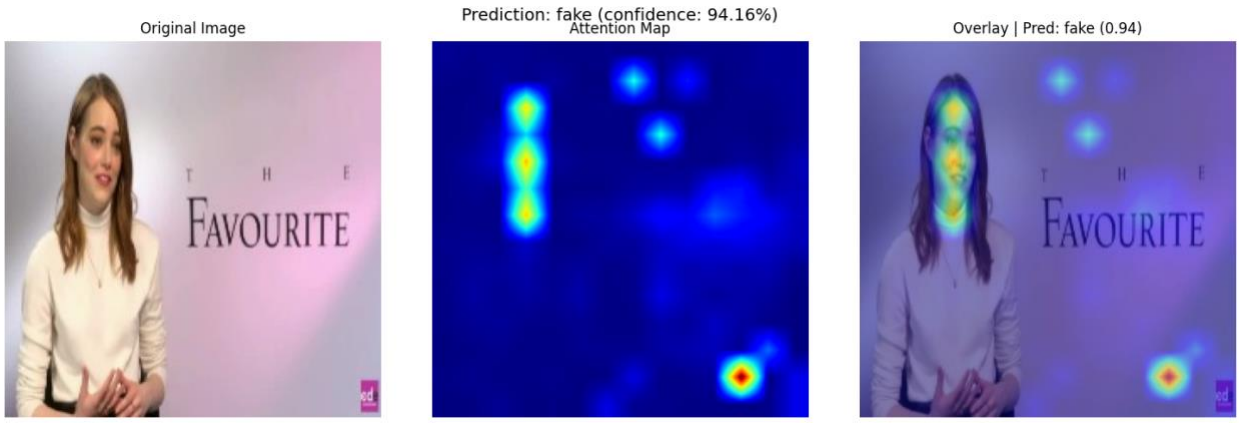

Figure 6-3. Attention maps for the top false positives (id48). Attention activates on background text and logos rather than facial features, indicating spurious correlation.

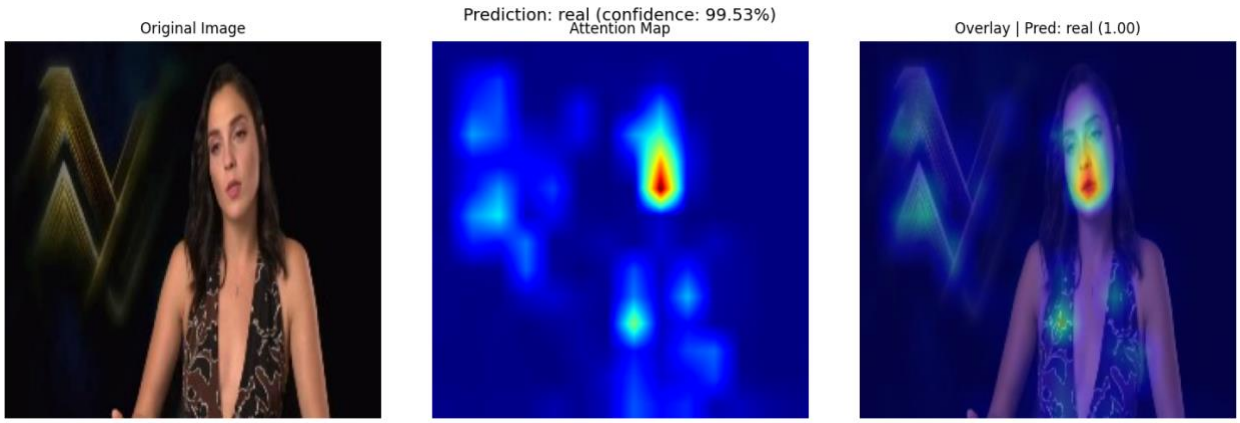

Figure 6-4. Attention maps for the top false negatives. Narrowly focused attention on a single region, lacking the distributed boundary scanning seen in successful detection.

The multi-layer attention visualization shows how the model builds its representation from early blocks to late blocks. In blocks 0 to 3 the attention is diffuse and global, covering the entire image including background, clothing, and the general scene. The model is performing low-level feature extraction at this stage, capturing edges, textures, and color patterns similar to what early CNN layers capture. By block 6 the attention begins narrowing to the subject, particularly the head and upper, body, with background

regions receiving less weight. By blocks 9 through 11 the attention is tightly concentrated on specific facial features. For correctly classified fake frames the attention in the final block remains distributed across facial boundaries. For correctly classified real frames it converges on a single area. For the false negative, the fake that fooled the model, the attention narrows much too quickly to a single point by block 6 and never develops the distributed scanning pattern. The model commits early to a localized region and misses the broader spatial inconsistencies that would have indicated manipulation.

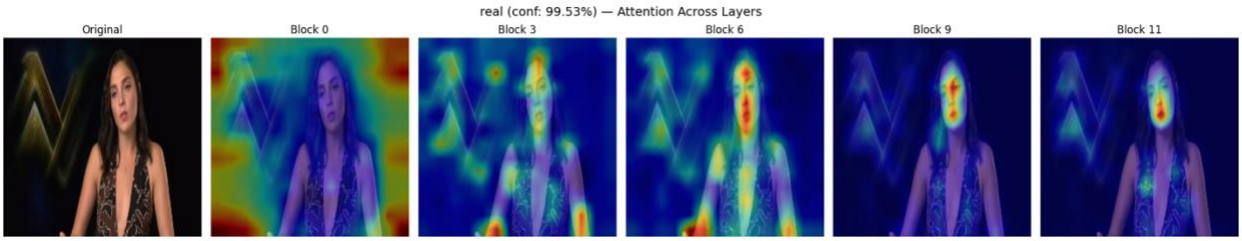

Figure 6-5. Attention evolution across transformer blocks 0, 3, 6, 9, and 11.

To move beyond individual examples, average attention maps were computed across 50 correctly classified samples from each class. The averaged fake attention map shows broad activation across the central face region covering the eyes, nose, mouth, and facial boundaries. The averaged real attention map is concentrated in a few small, localized regions. The difference map, showing where fake attention exceeds real attention, highlights the upper face around the forehead and hairline and the mid-face area. These are the regions where deepfake blending is most visible, the transition zone between generated and original content. The model appears to have learned to systematically examine these regions when evaluating whether a frame is synthetic. This confirms the model is not relying on arbitrary image features or dataset-specific

shortcuts. It has learned attention patterns that are consistent with domain knowledge about where deepfake artifacts appear, which validates the approach and makes the model's predictions interpretable by a human reviewer who understands facial manipulation.

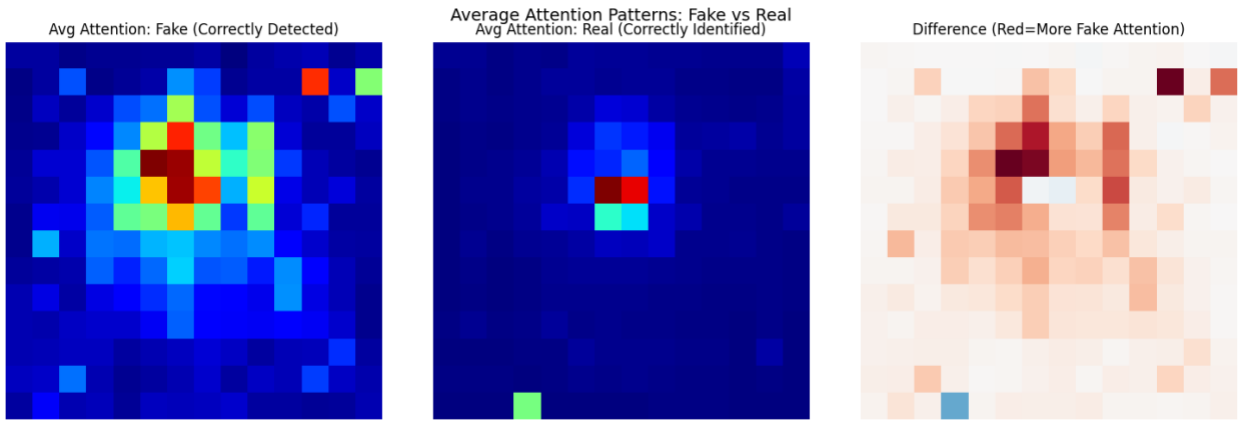

Figure 6-6. Average attention patterns across 50 correctly classified samples per class.

VII. <u>Error & Bias Analysis</u>

The false positive clustering around id48 in week 5 raised the question of whether errors are concentrated on specific individuals or spread evenly across the test set. Quantifying this across all identities reveals that the top 5 identities account for roughly 34% of all false positives. This is moderate concentration, meaning errors are partially identity-specific but more distributed than the top-confidence analysis initially suggested. The id48 dominance seen in Week 5 was an artifact of looking only at the most confident errors rather than the full population. For false negatives the errors spread across synthesis identities, reflecting which source individuals produce the most convincing fakes, those whose facial features are most faithfully reproduced by the synthesis algorithm. The identity concentration in false positives suggests the model

has learned identity-specific biases rather than a purely generalizable real versus fake distinction. Certain facial features, skin tones, or recording conditions associated with specific individuals trigger the model's fake indicators despite being authentic.

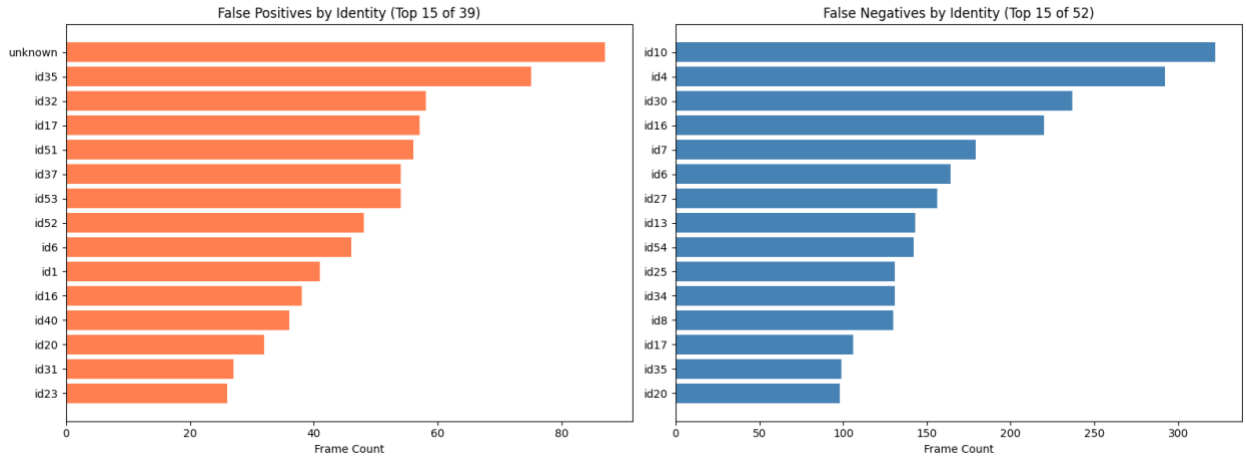

Figure 7-1. Error distribution by identity for false positives and false negatives.

Breaking errors down by source folder produces the most striking finding of the analysis. Celeb-real accuracy stands at 79.3% while YouTube-real accuracy is 97.0%. The model is significantly more likely to misclassify a Celeb-real image as fake than a YouTube-real image. The explanation for this disparity lies in the training data composition. Celeb-real and Celeb-synthesis share the same individuals. The model sees these celebrity faces far more often in the fake class than the real class and has learned to associate those faces with synthetic content. YouTube-real individuals do not appear in the synthesis set at all, making them easier to classify correctly as real because the model has no fake associations with those faces. This has direct implications for generalization. In a real-world deployment, the individuals most likely to be deepfakes are public figures and celebrities, precisely the population where this

model performs worst on authentic content. The model would disproportionately flag real content of the very people it is meant to protect.

Examining how confident the model is when it makes correct versus incorrect predictions reveals an asymmetry in its failure modes. Mean confidence for correct predictions is 0.7967 compared to 0.7752 for incorrect predictions, so the model generally assigns lower confidence to its mistakes. However, the dangerous failures are not the uncertain ones. There are 1,751 false negatives where the model assigns P(real) above 0.9, meaning it is highly confident that a fake image is real. These are the most dangerous failure cases because no threshold adjustment can recover them. In contrast only 2 false positives have P(real) below 0.1, so the model rarely makes confident mistakes in the fake direction. The asymmetry is significant for deployment. The model's primary failure mode is confidently passing fakes as real, which is the more dangerous direction for a deepfake detection system. A system designer would need to account for the fact that high confidence "real" predictions from this model are unreliable roughly 36% of the time, and the false negative danger has intensified with 1,751 confident misses.

The calibration curve reveals that the model’s predicted probabilities consistently overstate its actual reliability. When the model predicts P(real) of 0.6 the actual positive rate is only about 0.35. When it predicts P(real) of 0.95 the actual positive rate is approximately 0.82. The model is overconfident across nearly the entire probability range, with the most severe over confidence in the mid-range between 0.2 and 0.7. The prediction histogram shows an asymmetric bimodal distribution with a dominant peak near P(real) 0.15 to 0.20 and a secondary peak near 0.95 to 1.0. The model rarely

produces uncertain predictions around the 0.5 mark, instead committing to one class or the other with misplaced certainty. Platt scaling was applied as a post-hoc calibration method by fitting a logistic regression on the validation set's raw probabilities. However, this exposed the validation/test distribution mismatch concretely. The calibrator learned from the validation set that the base rate of real is approximately 13%, but the test set's actual base rate is approximately 35%. As a result the calibrated probabilities were pushed almost entirely below 0.5, predicting every single frame as fake at the default threshold. The AUC remained identical confirming that Platt scaling preserves ranking, but the probability remapping was useless for threshold-based decisions. This is not a failure of Platt scaling itself. Platt scaling is working exactly as designed, the failure is that the calibrator learned the wrong base rate because the validation set does not represent the test set's distribution. This finding transforms the validation/distribution mismatch from a theoretical concern raised in earlier weeks into a concrete, measurable failure mode. If it were to be deployed, any calibration method must be fit on data that matches the expected deployment distribution, as the default 0.5 threshold is too binary and can be catastrophically wrong when the distributions are shifted.

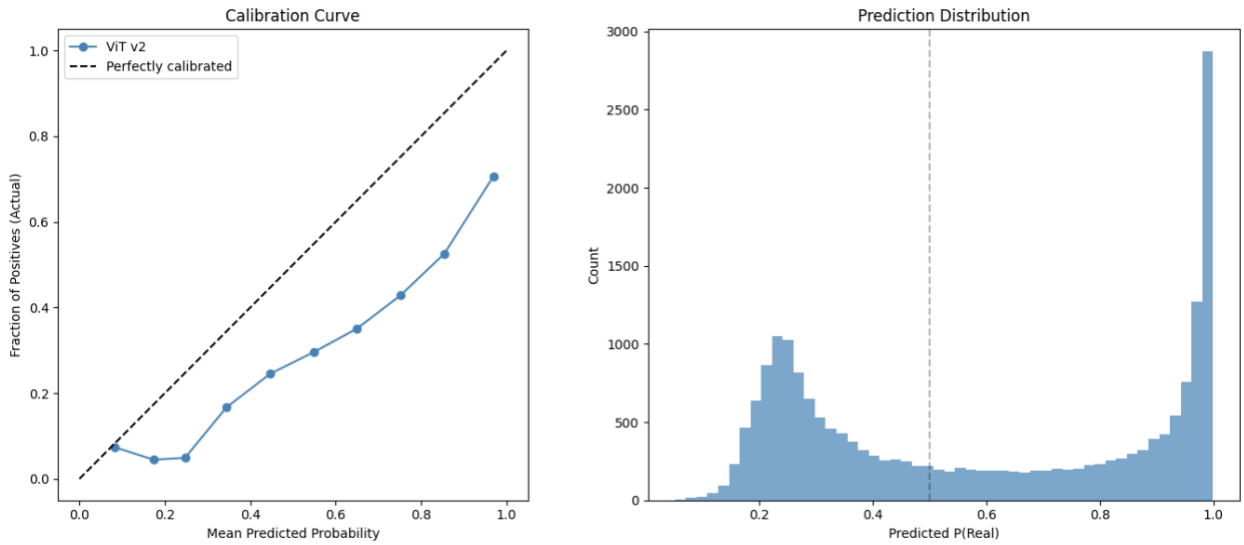

Figure 7-2. Calibration curve and prediction distribution.

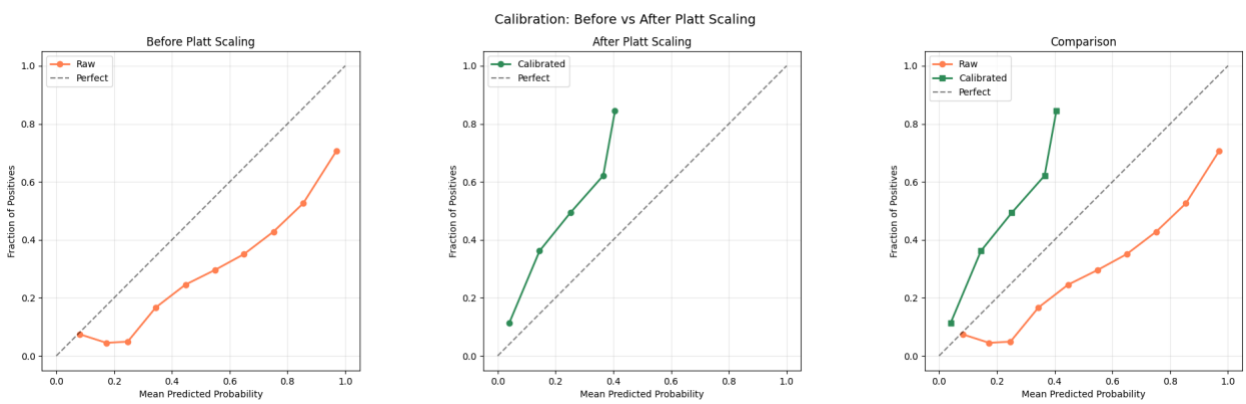

Figure 7-3. Calibration before and after Platt scaling.

To determine whether errors correlate with image quality rather than facial features, simple quality metrics were computed across all four prediction categories. The results show meaningful differences. False negatives, the missed fakes, are significantly darker with a mean brightness of 47.66 compared to 70 or higher for all other categories. The model struggles with low-light deepfakes, likely because the training data's fake examples are predominantly well-lit and the blending artifacts the model relies on are

obscured in darker images. False positives, the real images flagged as fake, have the lowest sharpness at 564.35 compared to true positives at 935.34 and the highest edge density at 0.1289. The combination of low sharpness and high edge density in misclassified real images suggests the model mistakes complex backgrounds, text overlays, or detailed facial features for manipulation boundaries. This is consistent with the Week 5 finding that attention sometimes latched onto background text. True positives, the correctly caught fakes, have the highest sharpness across all categories, which runs counter to the intuition that the model is simply detecting low quality as fake. It is genuinely identifying sharp, high quality deepfakes and catching them. The errors are not random with respect to image properties. They reflect systematic sensitivity to lighting and texture that could be addressed through targeted augmentation, specifically training with more low-light examples and images with complex backgrounds.

VIII. Robustness

The final evaluation tests how the model holds up when images are degraded, simulating the conditions deepfake content will encounter when circulating across a variety of social media. Three types of degradation were applied to the full test set without any retraining: JPEG compression at quality levels: 90, 70, 50, 30, and 10; Gaussian noise at sigma levels 5, 15, 25, and 50; and a downscale-upscale test that shrinks images to 112x112 then stretches them back to 224x224, Each perturbation was applied to every test frame and the model ran inference on the degraded versions using the v2 checkpoint. The baseline for comparison is the unperturbed test set performance with an AUC-ROC of 0.8409 and a macro F1 of 0.72.

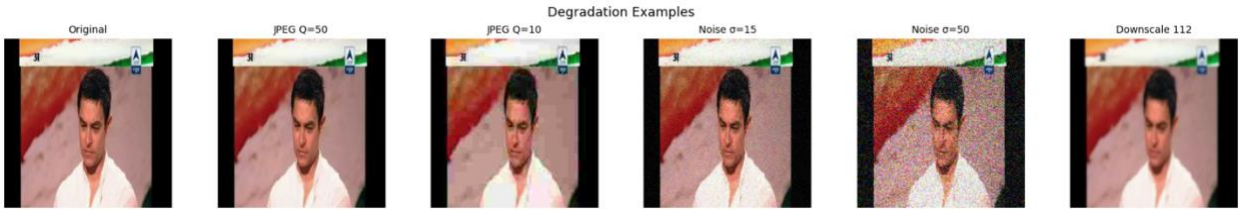

Figure 8-1. Visual examples of each degradation type applied to a sample frame.

Performance under compression dips in both AUC-ROC and macro F1 starting at Q=70 where AUC-ROC drops to 0.8179. The two metrics decline in parallel across the compression scale with the most severe degradation occurring at Q =10 where AUC-ROC falls to 0.7114 and macro F1 to 0.4955. At Q = 10 the model loses its discriminative ability and approaches random guessing. The model maintains acceptable robustness through Q = 50 where the drops remain moderate, but below Q = 50 performance degrades with increasing momentum. Since most social media platforms compress images to Q = 70 through 85, the model is reasonably robust in these circumstances for typical sharing conditions but would fail on heavily compressed or repeatedly reshared content.

Figure 8-2. Model performance under JPEG compression, Gaussian noise, and per-class recall.

Noise has more severe impact on the model's robustness than compression does. Unlike the compression test where AUC-ROC and macro F1 decline in parallel, under noise they diverge. Macro F1 shows a slight uptick to approximately 0.68 at sigma 5 then declines with the sharpest drop occurring between sigma 25 and sigma 50. AUC-ROC shows a more gradual stable decline across noise levels. This divergence combined with the sharp macro F1 decline indicates the model relies on fine-grained texture features that noise destroys. The model only maintains acceptable performance through sigma 15, after which it becomes too fragile for reliable detection. However, macro F1 never falls below 0.50, indicating the model retains some discriminative ability even under heavy noise rather than reverting entirely to random guessing.

The downscale-upscale test simulates the common scenario of content being screenshotted and reshared where resolution is lost in the process. AUC-ROC falls from 0.8409 to 0.7586, a substantial decrease of approximately 8%. Macro F1 declines more modestly. This indicates meaningful vulnerability to the resolution loss typical of screenshot pipelines where fine facial details that the model relies on for detection are blurred away by the resizing process.

Overall, in considering the model's robustness it would be viable under typical social media conditions where moderate compression and minimal noise are the norm. Most platforms compress within the Q = 70 to 85 range where the model maintains acceptable performance. However, the model cannot withstand the full range of image degradation it would encounter in unconstrained environments. Beyond certain degradation thresholds the manipulation features the model relies on for fake detection are destroyed, causing fake images to be misclassified as real at increasing rates. A

production system in the deployment scenario would need to account for the expected image quality in context and potentially flag images that fall below a quality threshold where the detection reliability breaks down. The robustness limitations combined with the calibration issues and identity bias findings from the error analysis paints a clear picture that this model can be a useful tool, but it needs to be human assisted under controlled conditions. Any thoughts of unsupervised, autonomous deployment are not going to go well at all.

IX. Limitations & Future Work

This project over the course of its time has had limitations come up that must be stated and discussed. The most significant was when fine tuning ViT v2, the lack of further ablation studies to fully see the changes, that were implemented simultaneously, isolated from each other for further sufficient evidence of the combination of changes being necessary. By only doing label smoothing, due to time constraints, the other three variables: augmentation, capacity, and GradScaler removal were never isolated to witness their effects. A full ablation gird testing on each change independently and in combination would have provide definitive attribution to the effects of each changing variables, yet within the timeline of this capstone along with computational cost made it impractical for full ablation to be done. The single ablation in existent is better than none at all but does not satisfy the question entirely.

The project relies on one single dataset. Celeb-DF-v2 is a well-established benchmark but it represents only one synthesis algorithm, one set of source identities, and one era of deepfake generation. The model was never tested on FaceForensics++, DFDC, or any other dataset, so its ability to generalize to different manipulation

techniques is completely unknown. The cross-dataset generalization would be more applicable on a larger timeline, so in current, the results speak only to performance on Celeb-DF-v2's specific manipulation style. In week 6 source-level bias was found, where Celeb-real accuracy was 79.3% versus YouTube-real at 97.0%, further underscores this limitation. The model has learned identity-specific associations between faces and the fake class that would likely transfer poorly to new individuals not seen during training.

The architecture also has its own limitations, as the project is limited to single-frame analysis. For a full video analysis, as would better match the dataset with it being video, it would have required ViT plus LSTM and or a lightweight temporal transformer for detecting frame-to-frame inconsistencies in the videos. Within the scope of the capstone this did not prove feasible within the computational resources and additional engineering complexity, but choosing single frame also meant that the current model cannot leverage temporal artifacts such as flickering, unnatural head movements, or inconsistent lighting across frames. These temporal cues could provide more reliable indicators of manipulation than spatial artifacts, this is because as generation continues to improve single frame spatial artifacts will become harder to detect. A natural extension would be aggregating predictions across multiple frames from the same video using even a simple mean or majority vote, which would provide some temporal robustness without the complexity and costs that come with sequence modeling.

The validation/test distribution mismatch is a recurring limitation throughout that affected model selection, metric interpretation, and calibration throughout the project. Training and validation sets have a 7:1 fake to real ratio while the test set is 1.8:1. This made validation metrics not directly comparable to the test metrics and caused the Platt

scaling calibration to fail when applied across the splits. In a production system the deployment distribution would likely differ from both, making this an unsolved problem. The model's probability outputs are not calibrated for any specific deployment context, and the 0.5 decision threshold is arbitrary. Future work should include calibrating on a held-out set that matches the expected deployment distribution or exploring distribution-agnostic calibration methods that do not assume a fixed base rate.

Several directions would strengthen the project if pursued beyond the capstone timeline. First, cross-dataset evaluation on FaceForensics++ would test whether the model's learned features transfer to different manipulation techniques or whether it has overfit to Celeb-DF-v2's synthesis algorithm. Second, identity-aware training that prevents the same individual from appearing both real and fake training splits to directly target the source-level bias that produced the 79.3% Celeb-real accuracy. Third, introducing temporal aggregation across frames, even without a dedicated sequence model, would provide a practical accuracy boost for video-level classification. Real-time inference optimization would be necessary for any deployment scenario, as the current ViT-base/16 is too large for edge devices or real-time video processing. Finally, the explainability work could be extended to include a human evaluation study comparing human moderator performance with and without attention map assistance, thereby directly testing the human-AI collaboration hypothesis from Groh et al. (2022).

X. Conclusion

This project set out to build a deepfake detection system that is not only accurate but interpretable, thus responding to a call from Groh et al. (2022) that detection systems would benefit from explainable AI rather than esoteric black box predictions. Over eight

weeks a complete pipeline was developed: dataset preparation, baseline CNN evaluation, Vision Transformer training and tuning, attention based explainability, structured failure analysis, and robustness testing. The progression from EfficientNet-B0 to ViT v1 to ViT v2 demonstrated consistent improvement in class-balanced detection, with macro F1 moving from 0.70 to 0.72, in v1 and v2, and AUC-ROC from 0.7706 to 0.8148 to 0.8409. The most practically meaningful result is real recall improving from 0.52 to 0.86 and false positives dropping by 72%, meaning the proportion of authentic content incorrectly flagged as fake is far less likely to be incorrectly flagged.

The project produced findings that go beyond just the performance numbers. The ablation experiment demonstrated, within its application, that no single regularization technique drove the improvement. Label smoothing alone shifted the model's class bias without improving its discrimination, proving that the v2 gains required the mutual combination of stronger augmentation, increased capacity, and label smoothing together. The attention maps validated that the model learned semantically meaningful detection patterns, focusing on facial boundaries and blending regions for fakes and targeted feature verification for reals, rather than exploiting dataset-specific shortcuts. The failure analysis revealed that errors are identity-specific and source-dependent, with Celeb-real accuracy at 79.3% versus YouTube-real at 97.0%, a finding with direct implications for deploying detection systems to protect public figures. The calibration analysis exposed the practical consequences of the validation/test distribution mismatch, demonstrating through the Platt scaling experiment that post-hoc calibration fails when fit on mismatched distributions. The robustness testing showed the model is

viable under typical social media compression but fragile against heavy noise and resolution loss, with fake recall degrading faster than real recall as image quality drops.

The attention maps are what distinguish this project from a standard classification exercise. The ability to visualize where the model looks when making a prediction transforms the system from a black box into a tool that a human moderator can question its decisions. A prediction of “fake” accompanied by a heatmap showing attention concentrated on facial blending boundaries gives the moderator evidence to evaluate rather than a number to blindly trust or reject. This is the core contribution; a detection system designed for human-AI collaboration where the model’s reasoning is visible and auditable. The findings from Groh et al. that inaccurate model predictions drag human accuracy down are less of a problem when the human can see what the model attended to and judge for themselves whether the reasoning makes sense.

Deepfake technology is going to continually improve. The synthetic algorithms generating the content will get better, the artifacts will get more subtle, and the line between real and generated content will continue to thus blur. Detection systems cannot win this defacto arms race through pure accuracy alone. They need to be interpretable so that the humans making final decisions can understand why a piece of content was flagged, transparent about their limitations so that deployment contexts match the model’s capabilities, and honest about what they get wrong so that failure modes are anticipated rather than discovered in production. This project is one step in that direction, not a full-on solution to the deepfake problem. But it does demonstrate that building detection systems with explainability at their core, rather than as an afterthought, produces tools that are more trustworthy. Which in a domain defined by

how successful deception is, trust in the detection system matters as much as its performance on a test set.

XI. Sources Cited

Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., & Fei-Fei, L. (2009). ImageNet: A large-scale hierarchical image database. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 248–255).

Groh, M., Epstein, Z., Firestone, C., & Picard, R. (2022). Deepfake detection by human crowds, machines, and machine-informed crowds. *Proceedings of the National Academy of Sciences*,https://www.pnas.org/doi/epdf/10.1073/pnas.2110013119

Li, Y., Yang, X., Sun, P., Qi, H., & Lyu, S. (2020). Celeb-DF: A large-scale challenging dataset for DeepFake forensics. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 3207–3216).